

Tanay Matta

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EDUCATION

Birla Institute of Technology and Science Pilani, Dubai Campus

Bachelors of Engineering in Computer Science, CGPA: 9.35

Dubai, UAE

Sept. 2022 – Present

TECHNICAL SKILLS

Focus: Generative AI, Agentic Systems, Applied Machine Learning, Data Engineering

Languages: Python, Golang, TypeScript, JavaScript, SQL, C

AI & ML Frameworks: LangChain, LangGraph, PyTorch, Ultralytics, Scikit-learn, Hugging Face, Foundational LLM(OpenAI, Anthropic)

Backend & APIs: FastAPI, Gin, Express.js, WebSocket, gRPC, RESTful Services

Data & Storage: BigQuery, Pinecone, MongoDB, Supabase, PostgreSQL, Redis, ETL Pipelines

DevOps & Cloud: Docker, Kubernetes, GCP, AWS, CI/CD, Render

Analytics & Tools: Pandas, NumPy, OpenCV, librosa, NLTK, Matplotlib, MLFlow, Langfuse

EXPERIENCE

Software Development(AI) Intern

Feb. 2026 - Present

ESRI Global

Sharjah, UAE

- Architected multi-agent invocation support for the ArcGIS RFT AI Assistant, enabling a single query to concurrently dispatch to multiple specialized agents, eliminating a core bottleneck and expanding the system's analytical capabilities.
- Engineered a context-management pipeline for the RFT AI Assistant, reducing token consumption by ~18% (110k → 90k tokens) through optimized prompt pruning and persistent chat history across conversation turns.
- Built and integrated an assistant chat thinking panel into the function-editor, implementing real-time polling, markdown rendering, and overflow handling to improve response transparency and UI stability.

Software Development Intern

Jul 2023 - Sept 2023 & May 2024 - Sept 2024

YMT Ads

Bengaluru, India (Remote)

- Designed and implemented automated ETL pipeline using Python and BigQuery API to extract and load large-scale datasets from Google Cloud Storage to local infrastructure, processing 2TB daily data volumes and reducing manual data retrieval workflows by 85%.
- Migrated existing infrastructure from PHP to Python, resulting in a 4-hour reduction in data retrieval time and significantly improving system performance.

Project Intern

Jun. 2024 - Aug. 2024

KPTAC Technologies

Dubai, UAE

- Optimized and streamlined data management processes for a proprietary platform by implementing a data migration pipeline from Excel to a proprietary backend system using a script written in Python, facilitating the automatic generation of menus and reducing manual input errors.

PROJECTS

HybridLM Engine | *Golang, GolangChain, OpenAI API, Groq, ONNX-Go, Gin, Redis, Docker*

- Engineered a hybrid LLM/SLM inference engine in pure Golang that dynamically routes requests between cloud-based LLMs and edge-deployed SLMs, reducing latency and compute costs for real-time applications.
- Orchestrated multi-provider LLM calls across OpenAI and Groq using GolangChain, with ONNX-Go for edge model execution, Redis caching for repeated queries, and Gin REST API containerized via Docker.

CrackIt | *Python, FastAPI, WebSocket, OpenAI Realtime API, Hugging Face Transformers, GPT-4o-mini, librosa*

- Built a real-time AI interview system with a WebSocket relay pipeline enabling live speech-to-speech conversation with adaptive follow-up question generation.
- Designed a multi-model evaluation pipeline analyzing candidate responses across semantic relevance (BERT), emotional state (Wav2Vec2 + DistilRoBERTa), and vocal delivery (librosa), with scores driving real-time question adaptation via LLM inference.

KanGL | *Python, LangGraph, LangChain, OpenAI API, LangSmith, FastAPI, Docker*

- Built a multi-agent AI system using LangGraph that autonomously scans codebases, detects frameworks, and generates deployment configurations via structured LLM reasoning pipelines.
- Architected a modular agent graph with tool-use, filesystem traversal, and token-optimized context extraction, deployed on Railway with LangSmith observability for tracing and cost monitoring.

Plgrzr | *Python, NLTK, Generative AI, Flask, OpenCV, Git*

- Developed a PDF document analysis system using NLTK, BERT embeddings, Google Cloud Vision API, and Mathpix to detect similarities in text content, handwriting patterns, and document structure with 95%+ accuracy in content extraction and semantic analysis.

ArsiDet | *Python, ML, PyTorch, Streamlit, LLM, Git*

- Developed an Arabic Sign Language Detection system using YOLO object detection and transfer learning, achieving 97% accuracy in real-time gesture recognition with an interactive Streamlit interface enabling webcam-based detection and LLM-powered translation.

ACHIEVEMENTS & CERTIFICATIONS

DAFZA x Wellington Hackathon: UAE Hackathon Winner

Dubai Airport Free Zone Authority, 20000 AED

April, 2025

Dubai, UAE

Samsung Innovation Campus AI Cohort

Samsung Gulf & UAE National Program for Coders, Final Score: 92/100

Nov. 2024

Dubai, UAE

Impactathon 1.0: Inter-University Hackathon

ACM Bits Pilani Dubai, Finished 4th place out of 30 teams

Feb. 2025

Dubai, UAE

Specialization in Golang

UCI, Irvine

Aug. 2025

Data Analysis with IBM

IBM

Aug. 2023

Software Engineering Simulation with Goldman Sachs

Goldman Sachs

Sep. 2023

Google Bigquery and PostgreSQL for Data Analysis

Udemy

Aug. 2022

LEADERSHIP POSITIONS

Peer Mentor

Student Council BPDC

Dubai, UAE

- Guiding incoming students through university transition by providing academic support, campus resource navigation, and promoting student engagement.

Technical Head

BITS Tech Fest BPDC

Dubai, UAE

- Led technical operations for the biggest tech-fest in the university by coordinating hackathons, and technical events while managing technical infrastructure and volunteer teams.
- Worked on building the registration website and managed database migration and deployment configurations while ensuring secure and fast participant registration.

Ex-Officio

Linux Users Group Bits Dubai

Dubai, UAE

- Helped administer the college's server architecture, implementing robust user state management on Linux servers.
- Conducted in-depth workshops for juniors on key topics like Database Management Systems (DBMS) and Data Structures and Algorithms (DSA)
- Engaged in outreach programs introducing K-12 students to technology and open-source principles.
- Led a cross-functional team of 16 students in organizing technical events, employing agile methodologies for efficient project management.

General Secretary

Expressions - The Public Speaking and Literary Club

Dubai, UAE

- Delivered engaging talks at multiple college events, enhancing audience participation and engagement by clearly presenting ideas to gatherings of 100+ attendees.
- Collaboratively managed declamation events, coordinating logistics, scheduling, participant registrations, and judging criteria with a hard-working team.